

CS189A Capstone Microsoft Team Presentation

Team:

Microsophons

Team members:

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1. Problems

- Programmers need to code face to face
- Cost time and money to meet
- Github merging issues

❏ The Solution

- A real-time collaborative IDE in a browser
- Code together in browser
- Real-time chat communication

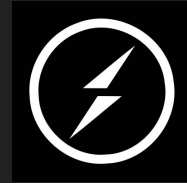
2. Technologies

→ Monaco

- Basic functions.

→ Socket io & Node js

- Communication



2. Technologies

→ Trello

- daily scrum
- tasks



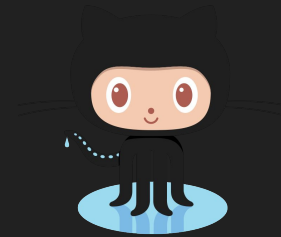
→ Cloud 9

- server



→ Github

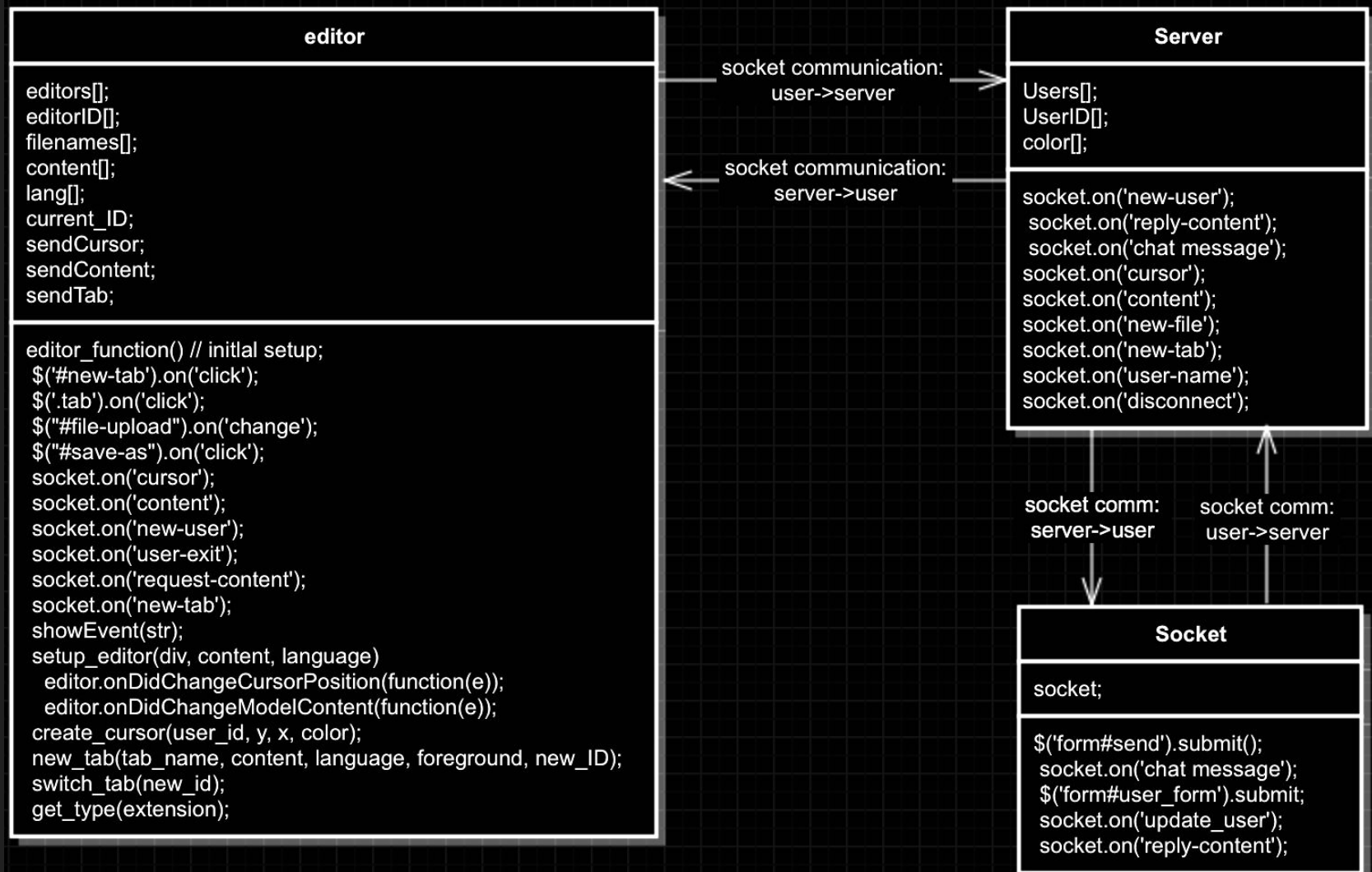
- management



3. Features

Current features supported:

- Cursor position synchronization
- Content change synchronization
- Upload local files to editor
- Save code to local files
- Tabs for multiple files
- Chat
- Show online users



3. Features: Cursor Sync example

On cursor position change event:

User -> server:

send 'cursor' signal, with userID, editorID, new cursor position,

Server -> other users:

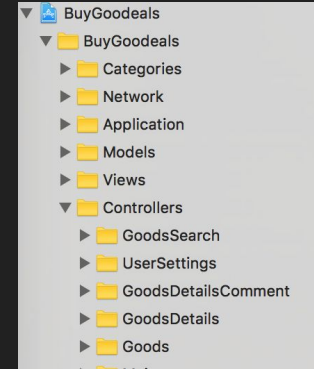
on 'cursor' signal : forward msg above + username, color

Other users: on 'cursor' signal:

- a) sendCursor = false;
- b) set cursor position for userID and editorID;
- c) Display tag with username and color
- d) sendCursor = true;

4. More features to be added

- Assemble a collection of files into a project
- Invite other people / set permissions 
- Import from/export to Github
- Create the database on Azure
 - File data/ Messages
- Display the output of the code in browser(If the code is web code:JS/html)



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5. Demo